The idea of positive and negative numbers

Numbers below zero — invented so that every subtraction has an answer.

LESSON 15–16 | 2 × 40 MIN | MATEMATIKA 7, §5 | S8 1

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

Why they are needed	15 min
Reading and writing	25 min
Direction as well as size	25 min
In practice	20 min
Homework	5 min

LEARNING OBJECTIVES

- Explain why negative numbers are needed.
- Read and write negative numbers, and use them in context.
- Interpret a negative number as a direction as well as a size.
- Solve simple problems involving temperature, debt and height.

TEXTBOOK

National	pp. 36–41
Cambridge	Stage 8, pp. 2–10
Workbook	Exercise 1.1

01 Explanation

Why they are needed

Among the numbers known so far, $3 - 5$ has no answer. Every other operation always works; only subtraction can fail.

OPERATION	ALWAYS POSSIBLE IN ℕ?
addition	yes
multiplication	yes
subtraction	**no** — $3 - 5$
division	no — $3 \div 5$ (that gave fractions)

Fractions were invented so that division always works; **negative numbers** are invented so that subtraction always works. The pattern is the same.

THE NUMBER SYSTEM GROWS TO FILL A GAP

Naturals, then fractions, then negatives, then irrationals in Grade 8, then complex numbers in Grade 11 — each new kind of number exists because some ordinary operation had no answer without it.

Reading and writing

WRITTEN	READ	MEANS
-5	minus five	five below zero
$+5$	plus five	five above zero
0	zero	neither positive nor negative
-0.4	minus nought point four	a negative fraction

The $+$ sign is usually left off: 5 and $+5$ are the same number. The $-$ sign is never left off.

ZERO IS NEITHER POSITIVE NOR NEGATIVE

It is the boundary between them, and it belongs to neither side. Statements such as “all numbers are positive or negative” are false because of zero.

Direction as well as size

A negative number carries two pieces of information: a **size** and a **direction**.

SITUATION	+	−
temperature	above zero	below zero
money	owned	owed
height	above sea level	below it
time	after an event	before it
movement	forwards	backwards

WHICH DIRECTION IS POSITIVE IS A CHOICE

Nothing makes “above sea level” positive; it is a convention. What matters is that once chosen it is kept — and that the two directions are opposite.

In practice

STATEMENT	AS A NUMBER
12 degrees of frost	$-12°C$
a debt of $50\ 000$ so‘m	$-50\ 000$
the Dead Sea shore, $430\ m$ below sea level	$-430\ m$
a lift going down 3 floors	-3
a loss of 7 points	-7

Example. The temperature is $-8°C$ and rises by 5 degrees. It becomes $-3°C$ — still below zero. It would need to rise by more than 8 degrees to reach the positive side.

> -12 IS COLDER THAN -5
>
> With negatives the bigger-looking number is the smaller one. Reading them on a thermometer, where -12 is plainly lower down, prevents the mistake.

02 Worked examples

EXAMPLE 1 **Write as numbers: 7 degrees of frost; a debt of $30\ 000$; $250\ m$ below sea level.**

1. Frost means below zero: $-7°C$.
2. A debt is money owed: $-30\ 000$.
3. Below sea level: $-250\ m$.
4. Each carries a direction as well as a size.

ANSWER

$-7, -30\ 000, -250$

EXAMPLE 2 **The temperature is $-8°C$ and rises by 5 degrees. Then it falls by 4. Find the final temperature.**

1. A rise of 5: $-8 + 5 = -3$.
   Still below zero.
2. A fall of 4: $-3 - 4$.
3. $= -7°C$
4. Colder than it began.

ANSWER

$-7°C$

EXAMPLE 3 **A lift starts on floor 5, goes down 8 floors, then up 2. Where does it stop?**

1. $5 - 8 = -3$
   Three floors below the ground.
2. $-3 + 2$
3. $= -1$
4. The first basement.

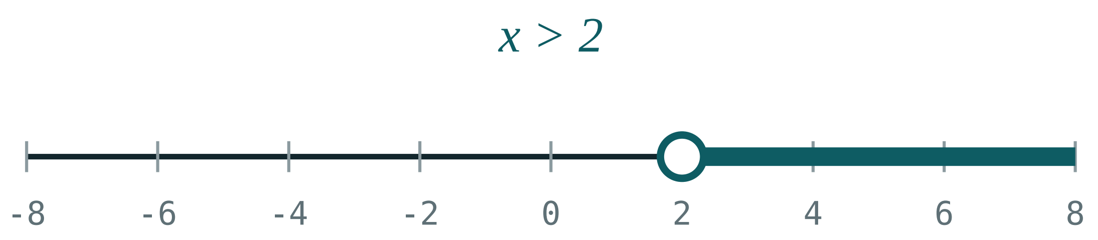
ANSWER

-1 — the first basement

03 Interactive model

Draw a thermometer up the side of the board and mark the day's temperatures; negative numbers arrive as readings, not as an abstraction.

Above and below zero



inequality **x > 2** | boundary **2 (open)** | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Positive number	Musbat son	Положительное число
Negative number	Manfiy son	Отрицательное число
Zero	Nol	Ноль
Sign	Ishora	Знак
Below zero	Noldan past	Ниже нуля
Debt	Qarz	Долг
Sea level	Dengiz sathi	Уровень моря
Rise and fall	Ko‘tarilish va pasayish	Подъём и спад

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Negative numbers were invented so that:

- addition works
- subtraction always works
- division works
- nothing

Zero is:

- positive
- negative
- neither
- both

7 degrees of frost is:

- $7°C$
- $-7°C$
- $70°C$
- $0°C$

A debt of $30\ 000$ is:

- $30\ 000$
- $-30\ 000$
- 0
- not a number

$-8 + 5$ equals:

- -13
- -3
- 3
- 13

Which is colder?

- $-5°C$
- $-12°C$
- equal
- cannot tell

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. 9 degrees of frost

   ANSWER -9

2. A debt of $20\ 000$

   ANSWER $-20\ 000$

3. $150\ m$ below sea level

   ANSWER -150

4. A gain of 6 points

   ANSWER $+6$

5. Is 0 positive?

   ANSWER No

6. Is 0 negative?

   ANSWER No

7. Colder: -3 or -9

   ANSWER -9

Medium · the standard question

7 PROBLEMS

1. -8 rises by 5

   ANSWER -3

2. -3 falls by 4

   ANSWER -7

3. Lift: floor 5, down 8, up 2

   ANSWER -1

4. Temperature -15 rises by 20

   ANSWER $+5$

5. A balance of $40\ 000$ after spending $55\ 000$

   ANSWER $-15\ 000$

6. Higher: $-200\ m$ or $-50\ m$

   ANSWER $-50\ m$

7. A submarine at $-80\ m$ rises $35\ m$

   ANSWER $-45\ m$

Hard · stretch and reason

7 PROBLEMS

1. The temperature falls from $4°C$ to $-11°C$: the fall

   ANSWER 15 degrees

2. A mountain $3200\ m$ above a lake $-150\ m$: the difference

   ANSWER $3350\ m$

3. Five readings $-3, 0, 2, -5, 1$: the coldest and the warmest

   ANSWER -5 and 2

4. Their range

   ANSWER 7

5. A lift from -2 to 7: floors travelled

   ANSWER 9

6. A debt of $80\ 000$ reduced by $95\ 000$

   ANSWER $+15\ 000$

7. Why is zero neither positive nor negative?

   ANSWER It is the boundary between them

07 Homework

Homework — 5 tasks

Give every answer with its sign and, where there is one, its unit.

1. Write as numbers: 14 degrees of frost; a debt of $45\ 000$; $60\ m$ below sea level.
2. The temperature is $-6°C$ and rises by 9 degrees. Find it now.
3. A lift starts on floor 3, goes down 7, then up 5. Where does it stop?
4. Which is higher: $-120\ m$ or $-45\ m$?
5. The temperature falls from $2°C$ to $-13°C$. By how many degrees?